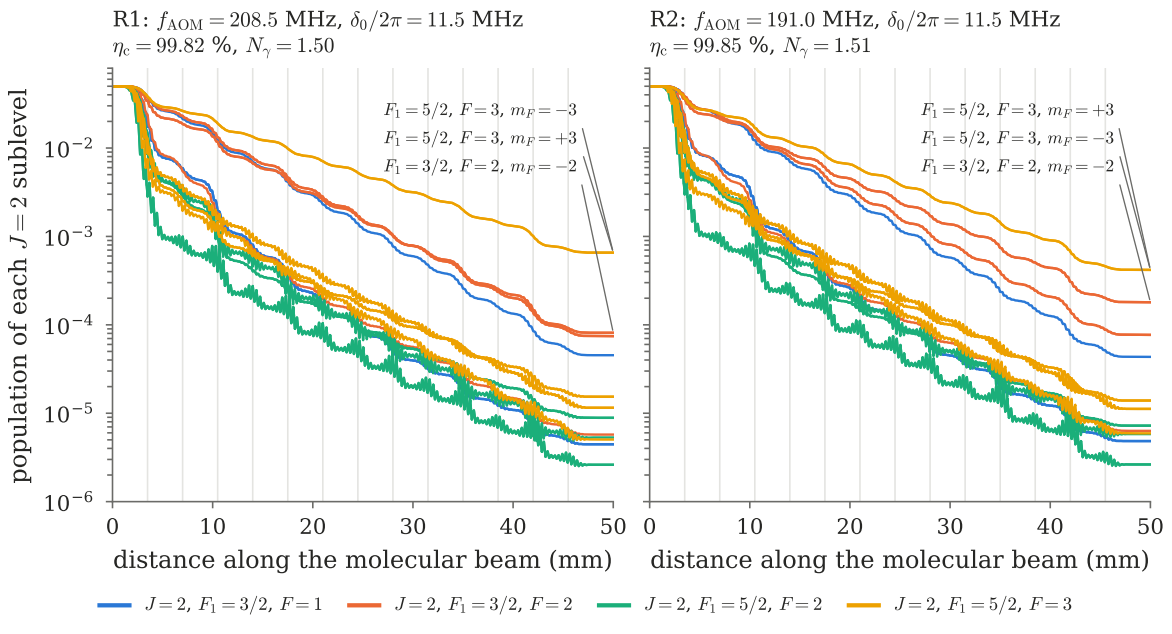

R1: $f_{\mathrm{AOM}} = 208.5$ MHz, $\delta_0/2\pi = 11.5$ MHz
$\eta_c = 99.82$ %, $N_\gamma = 1.50$
R2: $f_{\mathrm{AOM}} = 191.0$ MHz, $\delta_0/2\pi = 11.5$ MHz
$\eta_c = 99.85$ %, $N_\gamma = 1.51$
$F_1 = 5/2,\ F = 3,\ m_F = -3$
$F_1 = 5/2,\ F = 3,\ m_F = +3$
$F_1 = 3/2,\ F = 2,\ m_F = -2$
$F_1 = 5/2,\ F = 3,\ m_F = +3$
$F_1 = 5/2,\ F = 3,\ m_F = -3$
$F_1 = 3/2,\ F = 2,\ m_F = -2$
population of each $J = 2$ sublevel
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
0
10
20
30
40
50
distance along the molecular beam (mm)
distance along the molecular beam (mm)
$J = 2,\ F_1 = 3/2,\ F = 1$
$J = 2,\ F_1 = 3/2,\ F = 2$
$J = 2,\ F_1 = 5/2,\ F = 2$
$J = 2,\ F_1 = 5/2,\ F = 3$